

IoT and Digitalization for the Circular Economy

Lecture 0: Organization

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M.Sc Chintan Patel

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- Updated versions of these slides will be available in our [Github repository](#).

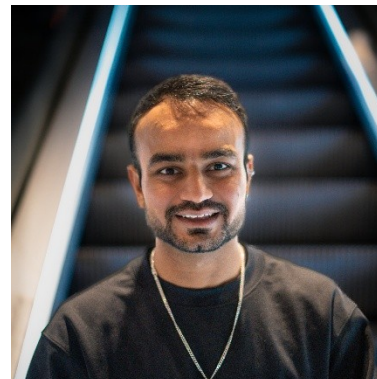
Team



Prof. Dr. Benjamin Leiding



M.Sc. Anant Sujatanagarjuna



M.Sc. Chintan Patel

Research Group

- **Emerging Technologies for the Circular Economy → ETCE**
- Research focus:
 - Intersection of IT and sustainability
 - Circular Economy and Circular Societies
 - Self-organized, decentralized and distributed systems
 - Sustainable and resilient food production
- Other courses:
 - Requirements Engineering (WS – M.Sc.)
 - The Limits to Growth – Sustainability and the Circular Economy (WS – open for everyone)



- ETCE Website – [Link](#)
 - Course material
 - Theses/project topics

Learning Outcome

- Understanding the concept of a circular economy, sustainability, and related concepts (biocapacity, etc.).
- Gain a basic understanding of causes, dimensions, and the characterization of climate change, environmental pollution, and dwindling non-renewable resources.
- Introduction to IoT and cyberphysical systems in the circular economy
- Sensors and actuators for IoT, control and process systems of the circular economy
- Experience in prototyping IoT applications and systems
- The ability to critically assess upcoming technological solutions enabling/facilitating sustainability and the circular economy.

Lectures

- 27.04.2026 → Organization (L00)
- 04.05.2026 → Introduction (L01)
- 11.05.2026 → Circular Economy (L02)
- 18.05.2026 → Lifecycle Assessment – LCA I (L03)
- 01.06.2026 → Lifecycle Assessment – LCA II (L04)
- 08.06.2026 → Introduction to the Internet of Things (L05)
- 15.06.2026 → Internet of Things – Communication (L06) + Security and Privacy (L07)
- 22.06.2026 → Internet of Things – Data Processing and Big Data (L08)
- **29.06.2026 → IoT in Mining I (L09)**
- **06.07.2026 → IoT in Mining II (L10)**
- 13.07.2026 → Digital Technologies and Greenwashing (L11)
- **23.07.2026 → Coding Workshop I (Goslar)**
- **24.07.2026 → Coding Workshop II (Goslar)**
- 27.07.2026 → Exam Q&A

Exercises (Deadlines)

- 18.05.2026 → Exercise 01 – Circular Economy (MC)
- 01.06.2026 → Exercise 02 – Lifecycle Assessment (LCA) – Part 1
- 08.06.2026 → Exercise 03 – Lifecycle Assessment (LCA) – Part 2
- 13.07.2026 → Exercise 04 – IoT in Mining
- 20.07.2026 → Exercise 05 – TBA

Course Organization

- News and updates → StudIP
- Organization of the lecture:
 - Slides are available on Github ([Link](#))
 - Please report bugs!
 - Lectures and exercises as live stream (BBB – next slide)
 - Lecture recordings will be available on StudIP and on Github
 - Exercise time slots = Time for questions and eventual tutorials related to the exercises

Questions? Write us an email: etce-iot@tu-clausthal.de ← **We will only respond to emails written to this specific email address!**

Dates/Times/Locations

- Lecture:
 - Monday **2:15 pm to 3:45 pm** (Berlin time) – **27.04.2026 to 13.07.2026**
 - Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton ([Link](#)) **OR video recordings**

- Exercise / Q&A:
 - Monday **4 pm to 4:45 pm** (Berlin time) – **27.04.2026 to 13.07.2026**
 - Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton ([Link](#))

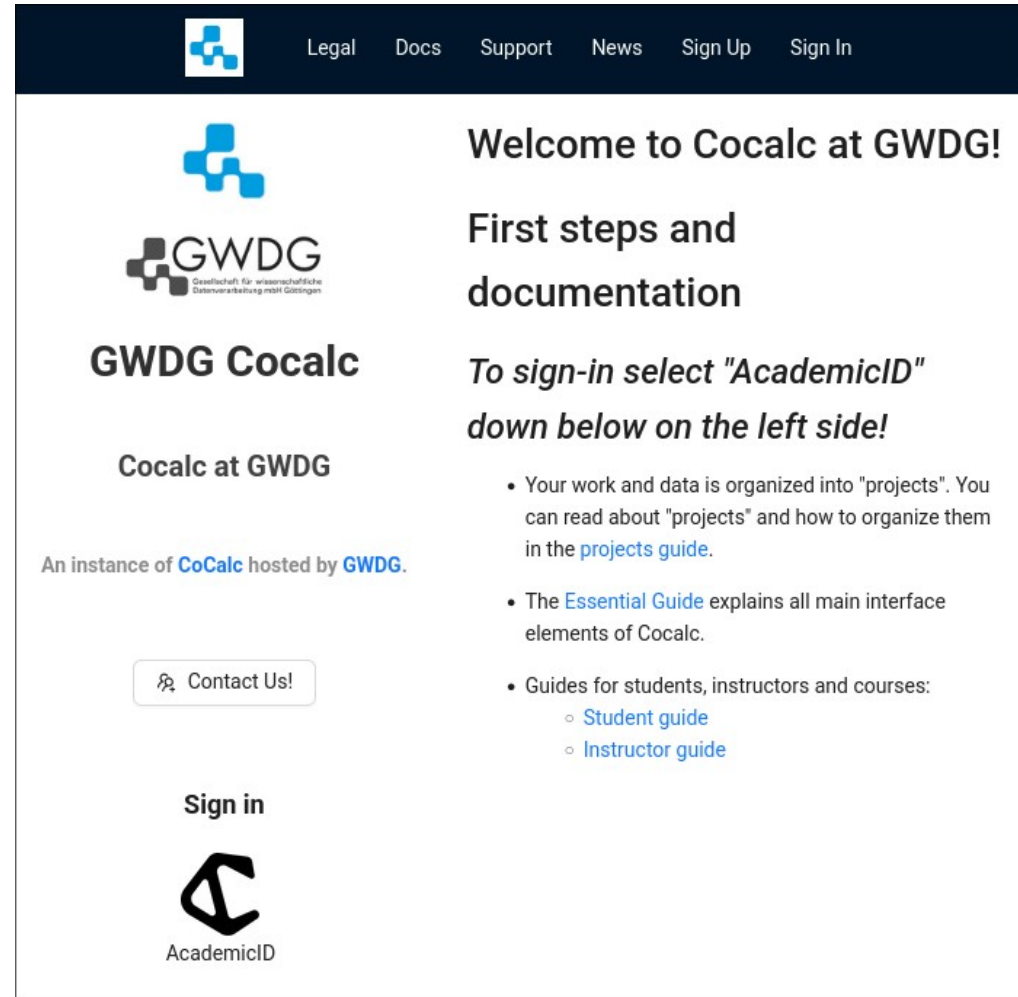
- Practical Coding Workshop:
 - When: **23.07.2026 and 24.07.2026** from **10 am - 5 pm (Berlin time)**
 - Location: Goslar (DIGIT – Wallstr. 6, 38640 Goslar)

Exercises and Practical Workshop

- Individual work → no group submissions
- Multiple-Choice or written exercises
- Multiple-Choice exercises are considered “self-evaluations” and are not submitted
- Submission of **each** written exercise is **mandatory**
- **You pass by receiving a grade greater than 0.**
- You will receive feedback on your submission (during Q&A session)
- Exercise = learning feedback
- Practical workshop → You pass the workshop if you score 50% (or more)

Exercise Submission and Grading

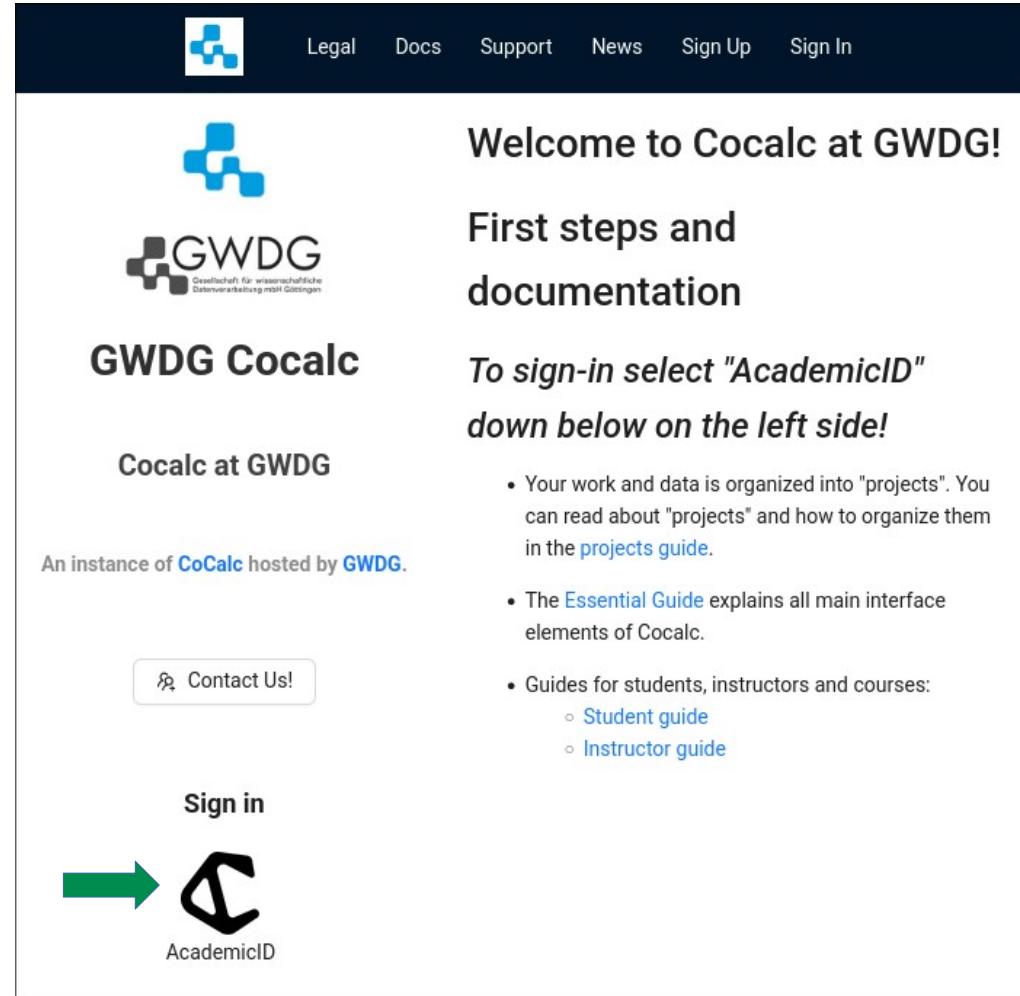
- Exercises are graded semi-automatically. Due to this it is highly important that you follow the required submission format. Otherwise the grading process will fail and you will receive 0 points.
- Exercises will be provided to you via Cocalc <https://cocalc.gwdg.de/> with additional instructions provided via [GitHub](#).



The screenshot shows the homepage of Cocalc at GWDG. The header is dark blue with a logo and navigation links: Legal, Docs, Support, News, Sign Up, and Sign In. The main content area is white. On the left, there is a blue logo, the GWDG logo (Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen), and the text "GWDG Cocalc". Below this, it says "Cocalc at GWDG" and "An instance of CoCalc hosted by GWDG." There is a "Contact Us!" button. At the bottom left, there is a "Sign in" button and the AcademicID logo. On the right, there is a welcome message: "Welcome to Cocalc at GWDG!" followed by "First steps and documentation". Below this, it says "To sign-in select 'AcademicID' down below on the left side!". There is a list of bullet points: "Your work and data is organized into 'projects'. You can read about 'projects' and how to organize them in the [projects guide](#)." "The [Essential Guide](#) explains all main interface elements of Cocalc." "Guides for students, instructors and courses:" with sub-points "Student guide" and "Instructor guide".

Coding Exercise Submission and Grading

- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo



The screenshot shows the homepage of Cocalc at GWDG. The top navigation bar includes links for Legal, Docs, Support, News, Sign Up, and Sign In. The main content area is divided into two columns. The left column features the GWDG logo, the text 'GWDG Cocalc', 'Cocalc at GWDG', and 'An instance of CoCalc hosted by GWDG.' Below this is a 'Contact Us!' button. At the bottom of the left column, there is a 'Sign in' section with a green arrow pointing to the AcademicID logo. The right column contains a welcome message, 'First steps and documentation', and instructions to select 'AcademicID' for sign-in. A list of links for projects, guides, and student/instructor guides is provided.

Legal Docs Support News Sign Up Sign In

Welcome to Cocalc at GWDG!

First steps and documentation

To sign-in select "AcademicID" down below on the left side!

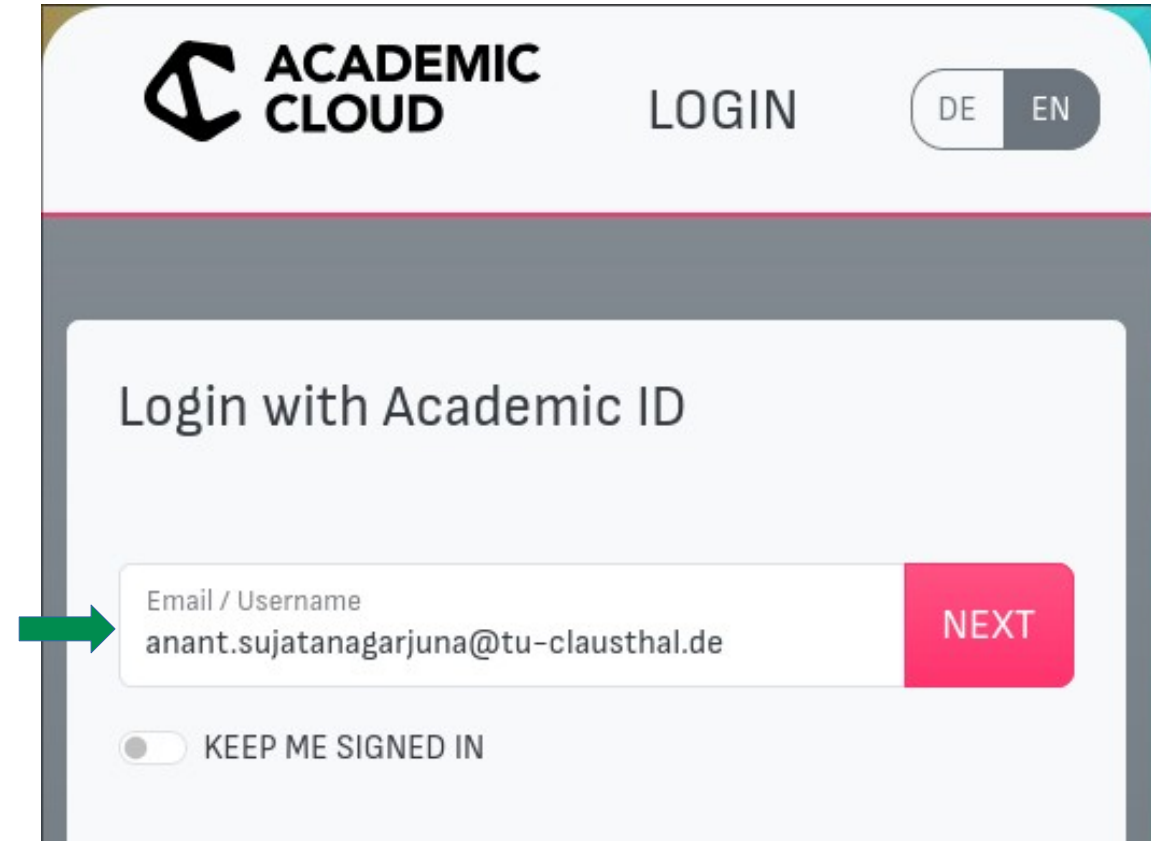
- Your work and data is organized into "projects". You can read about "projects" and how to organize them in the [projects guide](#).
- The [Essential Guide](#) explains all main interface elements of Cocalc.
- Guides for students, instructors and courses:
 - [Student guide](#)
 - [Instructor guide](#)

Sign in

AcademicID

Coding Exercise Submission and Grading

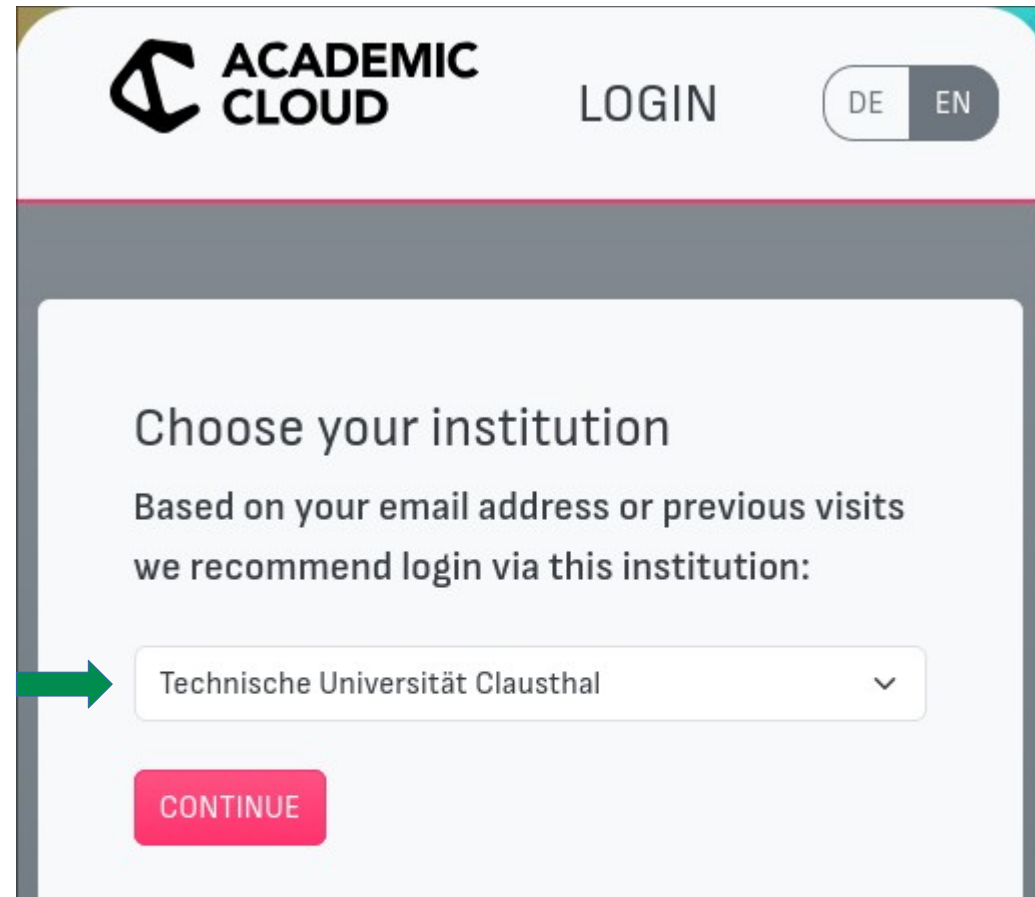
- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo
 2. Enter your university email address.



The image shows a screenshot of the Academic Cloud login interface. At the top, there is a header with the 'ACADEMIC CLOUD' logo on the left, a 'LOGIN' button in the center, and a language toggle switch on the right with 'DE' and 'EN' options. Below the header, the main content area is titled 'Login with Academic ID'. It features a large text input field labeled 'Email / Username' containing the email address 'anant.sujatanagarjuna@tu-clausthal.de'. A green arrow points to this input field. To the right of the input field is a red button labeled 'NEXT'. Below the input field, there is a toggle switch for 'KEEP ME SIGNED IN', which is currently turned off.

Coding Exercise Submission and Grading

- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo
 2. Enter your university email address.
 3. Choose “Technische Universität Clausthal” from the drop-down box.



ACADEMIC CLOUD LOGIN DE EN

Choose your institution

Based on your email address or previous visits we recommend login via this institution:

Technische Universität Clausthal

CONTINUE

Coding Exercise Submission and Grading

- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo
 2. Enter your university email address.
 3. Choose “Technische Universität Clausthal” from the drop-down box.
 4. Login with your usual TU-Clausthal credentials.



TU Clausthal
Anmelden bei Academic Cloud



Academic Cloud

Benutzername

Passwort

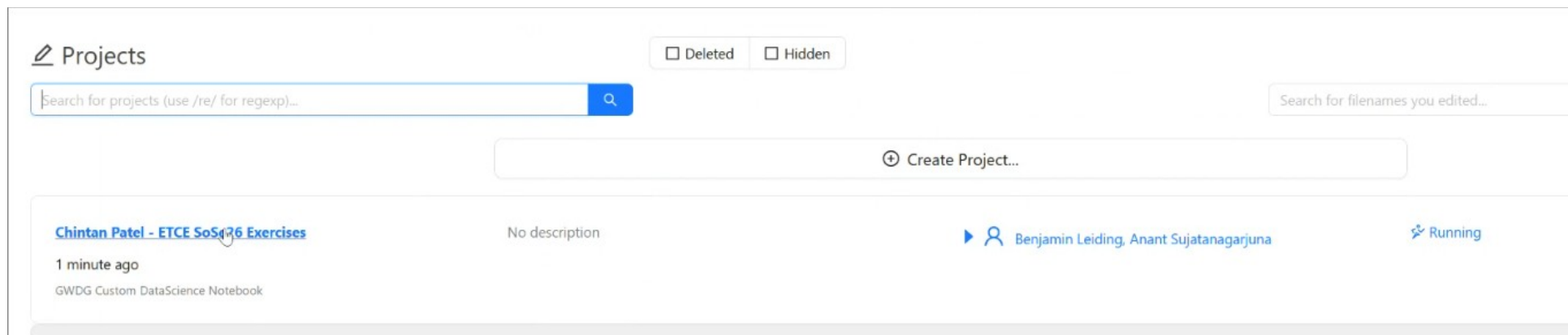
☐ Anmeldung nicht speichern

☐ Die zu übermittelnden Informationen anzeigen, damit ich die Weitergabe gegebenenfalls ablehnen kann.

Anmelden

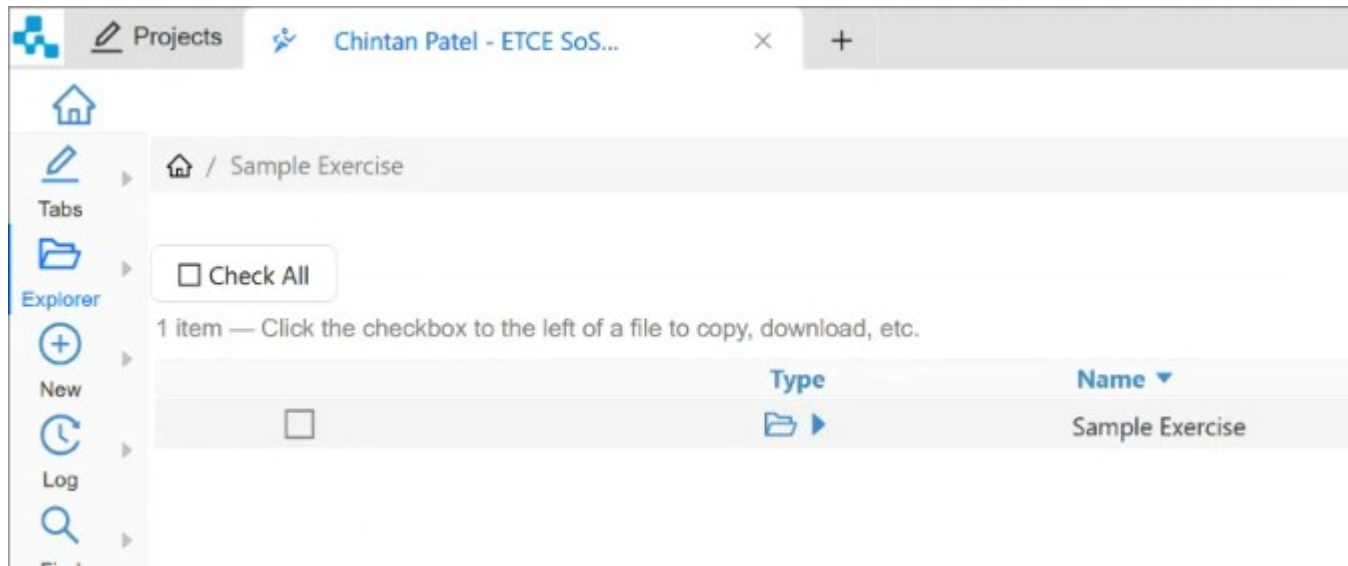
Coding Exercise Submission and Grading

- Submitting Assignments
 1. Click on “IoT Mining SoSe26”



Coding Exercise Submission and Grading

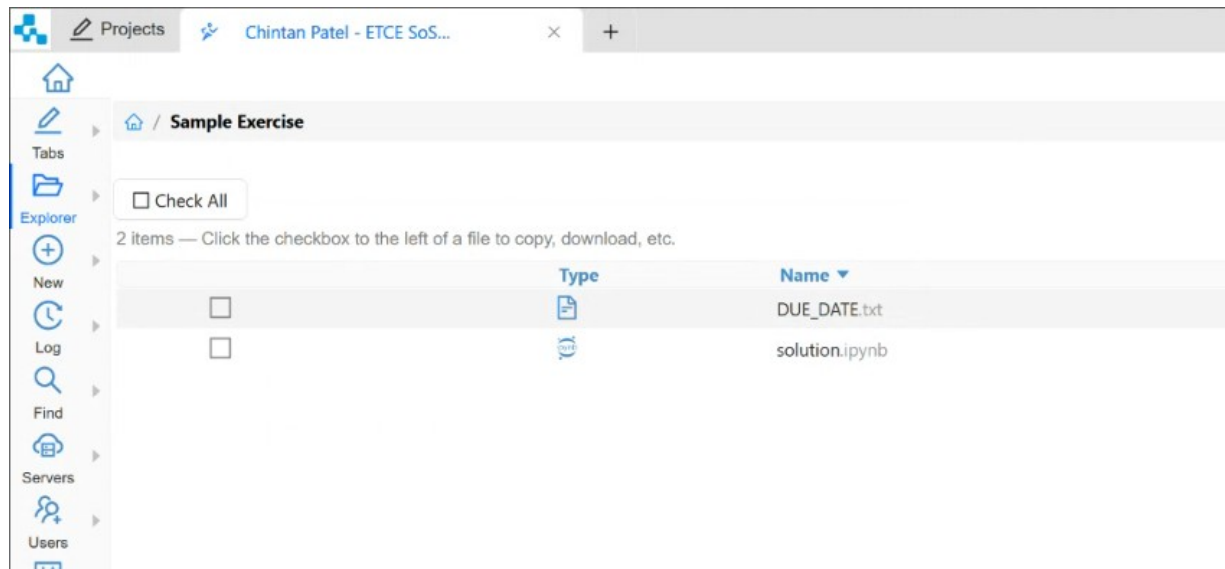
- Submitting Assignments
 1. Click on “IoT Mining SoSe26”
 2. Open the folder of the exercise you want to work on.



Coding Exercise Submission and Grading

■ Submitting Assignments

1. Click on “IoT Mining SoSe26”
2. Open the folder of the exercise you want to work on.
3. Edit the file in the exercise folder as indicated in the exercise handout PDFs.



Coding Exercise Submission and Grading

■ Submitting Assignments

1. Click on “IoT Mining SoSe26”
2. Open the folder of the exercise you want to work on.
3. Edit the file in the exercise folder as indicated in the exercise handout PDFs.



The screenshot shows a Jupyter Notebook interface with a sidebar on the left containing icons for Log, Find, Servers, Users, Upgrades, Processes, and Settings. The main area is titled "Sample Programming Exercise" and contains two code cells. The first cell, labeled "In [24]:", contains Python code for a function `get_the_date()` that returns the current date. The second cell, labeled "In [23]:", contains Python code for a function `evaluate()` that checks if the output of `get_the_date()` is a `datetime.date` object and returns a grade of 100 if it is, and 0 otherwise. The output of the second cell is "Out[23]: Your grade is: 100%".

```
Log
Find
Servers
Users
Upgrades
Processes
Settings

Sample Programming Exercise
Edit the code in the following code blocks to complete your submission for this programming assignment

In [24]:
import datetime
def get_the_date() -> datetime.date:
    """
    This function should return today's date.
    """
    pass

Once you are done writing your solution, you can evaluate your solution in the code cells below:

In [23]:
def evaluate():
    """
    Use this function to evaluate whether your code works, and to see what grade your code will receive.
    """
    if type(get_the_date()) == datetime.date:
        return 100
    else:
        return 0

    print("Your grade is: "+str(evaluate()))+"%")

Out[23]: Your grade is: 100%
```

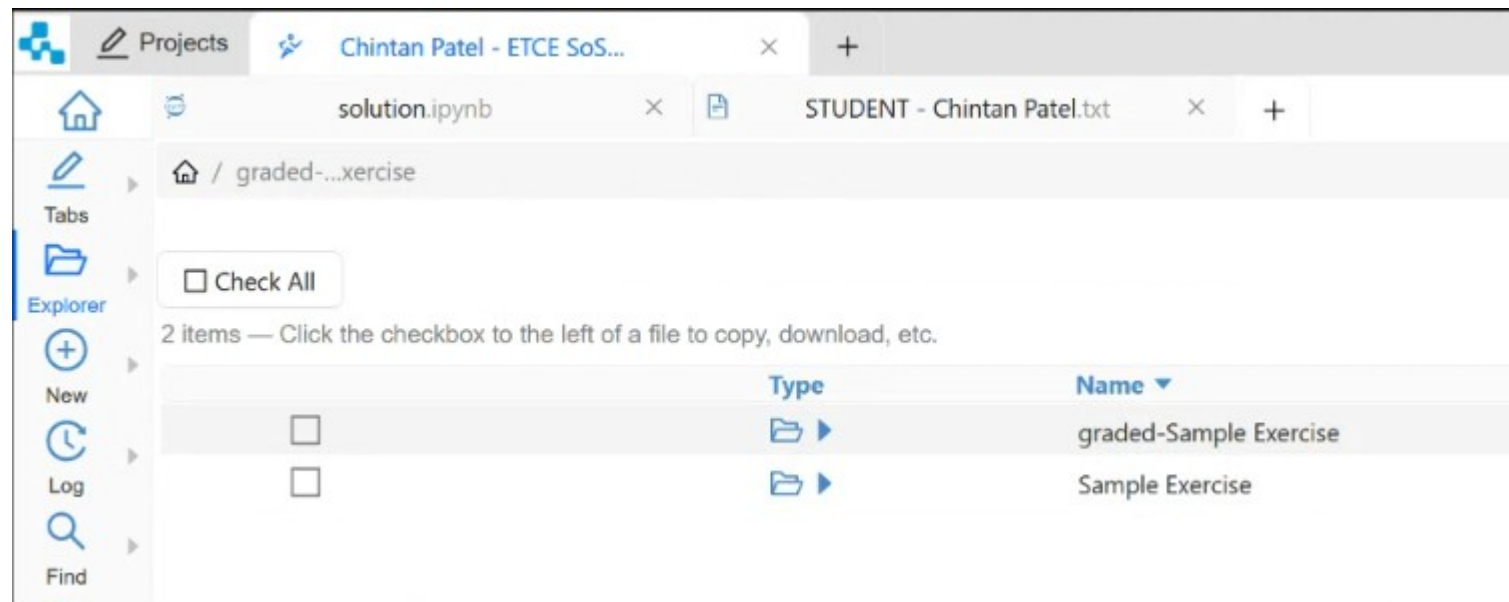
Coding Exercise Submission and Grading

- Exercise Grades
 - We will collect your exercises after the deadline has elapsed.
 - Please ensure to save all changes to the exercise folder BEFORE the deadline.
 - DO NOT submit your assignments via email/other means.

Coding Exercise Submission and Grading

■ Exercise Grades

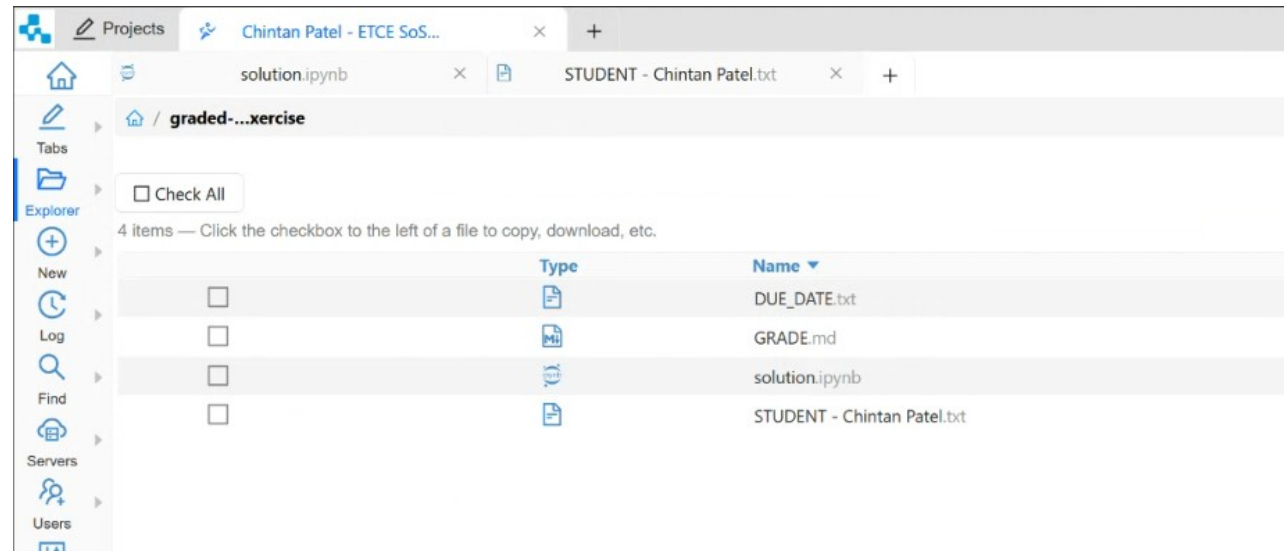
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- Exercises, once graded will be released back to you.



Coding Exercise Submission and Grading

■ Exercise Grades

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- Please ensure to save all changes to the exercise folder BEFORE the deadline.
- DO NOT submit your assignments via email/other means.
- Exercises, once graded will be released back to you.
- Your grade is provided inside the “GRADE.md” file.



Coding Exercise Submission and Grading

- Exercise Grades
 - We will collect your exercises after the deadline has elapsed.
 - Please ensure to save all changes to the exercise folder BEFORE the deadline.
 - DO NOT submit your assignments via email/other means.
 - Exercises, once graded will be released back to you.
 - Your grade is provided inside the “GRADE.md” file.
 - Receiving a grade of 0 = FAIL.

Multiple-Choice Exercises

Every student enrolled in this course is advised to take the knowledge quiz in first two weeks of the course.

- **Goal of the test:**
 - To check the knowledge level of the student that is relevant to this course of study.
- **Preparation:**
 - A review of basic concepts of Cryptography and Circular Economy is recommended for Week 1 and Week 2 respectively
 - Knowledge quiz for Week 1 only tests your existing knowledge.
- **THE TEST IS JUST FOR YOU → WE CANNOT CHECK THE TEST RESULTS**

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Link for all multiple-choice exercises → etce.etce-lab.de

Multiple-Choice

E01: Prior Knowledge

Introduction

This question is the exercise for "*L01 - Introduction*" of the "Emerging Technologies for the Circular Economy" course taught at the TU Clausthal. All the course material is hosted on our [GitHub](#) page. This multiple-choice quiz tests your prior knowledge of the topics that we will discuss during the course.

You must answer at least half **the questions** correctly to pass the exercise.

If you have any questions, you can contact us via e-mail: etce-etce@clausthal.de

Question 01: Block ciphers are used in symmetric cryptography to encrypt and decrypt data. They operate on blocks of a fixed length (usually 64 or 128 bits). To encrypt some data, you could split it into block sized chunks and simply encrypt them separately. What which of the following properties are true?

Choose one or more options then select Submit.

- ☐ This is called ECB mode
- ☐ This is called CBC mode
- ☐ Patterns in the data will show up, which can impact security
- ☐ There is no integrity protection
- ☐ It will be easy to calculate the key

Submit

Multiple-Choice Exercises

Question 01: Block ciphers are used in symmetric cryptography to encrypt and decrypt data. They operate on blocks of a fixed length (usually 64 or 128 bits). To encrypt some data, you could split it into block sized chunks and simply encrypt them separately. What which of the following properties are true?

Choose one or more options then select Submit.

This is called ECB mode

This is called CBC mode

Patterns in the data will show up, which can impact security

✓ There is no integrity protection

It will be easy to calculate the key

Show correct answer



Examination

- Prerequisite for admission to the final exam (all criteria have to be fulfilled):
 - Submit all exercises
 - Pass the practical workshop

- Final exam:
 - **Most likely 12.08.2026 and 13.08.2026**
 - Oral examination (20min)
 - Online

Self-Study Star

Self-Study Star → 

- Slides with the self-study star indicate optional/additional study material that is **not** mandatory but could be helpful or interesting

Literature

- This course is not based on a single book and you **do not** need to buy a book to pass the exam.
- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *The Limits to Growth* (1972).
- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *Limits To Growth: The 30-Year Update* (2004).
- Baccini et al. *Metabolism of the Anthroposphere: Analysis, Evaluation, Design* (2012).
- Walter R. Stahel. *The Circular Economy: A User's Guide* (2019).
- W. Brian Arthur. *The Nature of Technology: What It Is and How it Evolves* (2011)
- David Wallace-Wells. *The Uninhabitable Earth, Annotated Edition* (2017).
- (German) Stefan Rahmstorf, Hans Joachim Schellnhuber. *Der Klimawandel* (2019).

Literature

- Perry Lea. Internet of Things for Architects: Architecting IoT solutions by implementing sensors, communication infrastructure, edge computing, analytics, and security (2018).
- M.A. Khan, M.T. Quasim, F. Algarni, A. Alharthi. *Decentralised Internet of Things* (2020).
- Dimitrios Serpanos und Marilyn Claire Wolf. *Internet-of-Things (IoT) Systems Architectures, Algorithms, Methodologies* (2018).

Questions?